BATCH NO:M2001

DESIGN AND DEVELOPMENT OF SMART PLANT MICRO-MACRO NUTRIENT DOSER FOR DRIP IRRIGATION

Minor Project-II report submitted
in partial fulfillment of the requirement for award of the degree of

Bachelor of Technology
in
Computer Science and Engineering
(Artificial Intelligence and Machine Learning)

By

R.SAI PADMINI	(23UECL0051)	**(24646)**
V.HARSHITA	(23UECL0063)	**(25115)**
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Under the guidance of
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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
(ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING)
SCHOOL OF COMPUTING

VEL TECH RANGARAJAN DR. SAGUNTHALA R&D INSTITUTE OF SCIENCE AND TECHNOLOGY

(Deemed to be University Estd u/s 3 of UGC Act, 1956)
Accredited by NAAC with A++ Grade
CHENNAI 600 062, TAMILNADU, INDIA

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CERTIFICATE

It is certified that the work contained in the project report titled "DESIGN AND DEVELOPMENT OF SMART PLANT MICRO-MACRO NUTRIENT DOSER FOR DRIP IRRIGATION" by R.SAI PADMINI (23UECL0051), V.HARSHITA (23UECL0063), V.MASTHANAMMA (23UECL2003) has been carried out under my supervision and that this work has not been submitted elsewhere for a degree.

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DECLARATION

We declare that this written submission represents our ideas in our own words and where others' ideas or words have been included, we have adequately cited and referenced the original sources. We also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in our submission. We understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

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APPROVAL SHEET

This project report entitled DESIGN AND DEVELOPMENT OF SMART PLANT MICRO-MACRO NUTRIENT DOSER FOR DRIP IRRIGATION by R.SAI PADMINI (23UECL0051), V.HARSHITA (23UECL0063), V.MASTHANAMMA (23UECL2003) is approved for the degree of B.Tech in Computer Science and Engineering (Artificial Intelligence and Machine Learning).

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ABSTRACT

This project presents the design and implementation of a smart automated nutrient dosing system for modern agriculture. Traditional irrigation and fertilization methods are often inefficient and labor-intensive. They result in overuse or underuse of water and nutrients, affecting crop growth. Such practices also contribute to soil degradation and reduced agricultural productivity. To address these issues, an automated and intelligent system is proposed. The system is built using an ESP32 microcontroller for efficient control and processing. It integrates various sensors to monitor real-time environmental and soil conditions. Soil moisture sensors measure the water content in the soil. Temperature sensors help in understanding climatic conditions affecting crops. Flow sensors are used to monitor the water supply accurately. Based on sensor data, the system makes decisions automatically. A peristaltic pump is used to inject nutrients into the irrigation system. This ensures precise and controlled delivery of fertilizers. A MOSFET module is used to control high-power devices safely and efficiently. The nutrients are mixed with water using an injection tee mechanism. The mixed solution is delivered directly to plants through drip irrigation. This method ensures uniform distribution and minimizes wastage. The system significantly reduces manual intervention and labor costs. It improves water efficiency and optimizes fertilizer usage. As a result, plant growth and crop yield are enhanced. The system also helps in maintaining soil health over time. An optional IoT feature enables remote monitoring and control. Users can access real-time data through mobile or web applications. This adds convenience and improves decision-making for farmers. Overall, the system promotes sustainable and precision agriculture practices.

Keywords: Smart Irrigation, Nutrient Dosing System, Internet of Things (IoT), ESP32, Precision Agriculture, Soil Moisture Monitoring, Fertigation, Peristaltic Pump, Environmental Sensors, Sustainable Agriculture.

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LIST OF ACRONYMS AND ABBREVIATIONS

AC	Alternating Current
ADC	Analog to Digital Converter
AI	Artificial Intelligence
API	Application Programming Interface
ARIMA	AutoRegressive Integrated Moving Average
BT	Bluetooth
DAC	Digital to Analog Converter
DC	Direct Current
EC	Electrical Conductivity
ESP32	Espressif 32-bit Microcontroller with Wi-Fi and Bluetooth
GPIO	General Purpose Input Output
IoT	Internet of Things
LSTM	Long Short-Term Memory
ML	Machine Learning
pH	Potential of Hydrogen
PWM	Pulse Width Modulation
RF	Random Forest
USB	Universal Serial Bus
Wi-Fi	Wireless Fidelity

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Chapter 1

INTRODUCTION

1.1 Introduction

Modern agriculture faces significant challenges due to the increasing demand for food production, limited water resources, and the need for sustainable farming practices. Traditional irrigation and fertilization methods are largely manual and often inefficient, leading to improper utilization of water and nutrients**[1]**. Farmers frequently rely on fixed schedules or visual inspection, which can result in over-irrigation or under-irrigation, causing water wastage and negatively impacting crop health. Similarly, the uncontrolled application of fertilizers may lead to nutrient imbalance, soil degradation, and environmental pollution. These issues not only reduce crop yield and quality but also increase operational costs and labor dependency**[5]**. Therefore, there is a growing need for intelligent and automated systems that can optimize resource usage and enhance agricultural productivity.In recent years, advancements in embedded systems and Internet of Things (IoT) technologies have opened new possibilities for precision agriculture. Smart farming solutions enable real-time monitoring, data-driven decision-making, and automation of agricultural processes. This project presents the design and development of a smart automated nutrient dosing system integrated with drip irrigation[6]. The system utilizes an ESP32 microcontroller as the core processing unit, which offers built-in Wi-Fi and Bluetooth capabilities for efficient communication and control. Various sensors, including soil moisture sensors, temperature sensors, and flow sensors, are employed to continuously monitor environmental and soil conditions. The collected data is processed by the microcontroller to make intelligent decisions regarding irrigation and nutrient delivery**[2]**.

1.2 Background

Agriculture has been the backbone of human civilization, playing a vital role in food production and economic development. However, traditional farming practices have largely remained dependent on manual labor, experience-based decisions, and fixed irrigation schedules**[5]**. These conventional methods often fail to consider real-time environmental conditions such as soil moisture, temperature, and plant nutrient requirements. As a result, farmers frequently face issues such as over-irrigation, under-irrigation, and inefficient fertilizer usage. Overuse of water not only leads to wastage of a critical natural resource but also contributes to soil erosion and nutrient leaching. Similarly, excessive or insufficient application of fertilizers can disrupt soil composition, reduce fertility, and negatively impact crop yield. These challenges highlight the limitations of traditional approaches and the need for more efficient and sustainable solutions**[6]**.

With the rapid advancement of technology, modern agriculture is gradually shifting towards precision farming techniques. Precision agriculture focuses on optimizing resource utilization by using data-driven approaches and automated systems. The integration of embedded systems, sensors, and communication technologies has made it possible to monitor field conditions in real time and respond accordingly**[7]**. Microcontrollers such as the ESP32 have become popular due to their low cost, high performance, and built-in wireless capabilities. These features enable seamless connectivity and remote monitoring, making them suitable for smart agricultural applications. Sensors such as soil moisture sensors, temperature sensors, and flow sensors play a crucial role in collecting accurate data from the field, which can be used to make informed decisions regarding irrigation and nutrient supply.

1.3 Objective

The primary objective of this project is to design and develop a smart automated nutrient dosing system integrated with drip irrigation to improve agricultural efficiency and sustainability. The system aims to reduce manual intervention by automating irrigation and fertilizer application based on real-time sensor data. It focuses on optimizing the usage of water and nutrients to prevent wastage and ensure proper plant growth. By using an ESP32 microcontroller, the project seeks to enable intelligent decision-making and reliable control of irrigation processes. Another key objective is to monitor environmental conditions such as soil moisture, temperature, and water flow to maintain optimal growing conditions**[9]**. The system is also intended to deliver precise amounts of micro and macro nutrients using a controlled dosing mechanism. Additionally, it aims to minimize operational costs and labor requirements for farmers. The project supports the concept of precision agriculture by improving crop yield and maintaining soil health. Integration of IoT features is considered to allow remote monitoring and control through digital platforms. Finally, the objective is to develop a cost-effective, scalable, and user-friendly solution that can be easily implemented in real-world farming environments**[10]**.

1.4 Problem Statement

Agriculture continues to rely heavily on traditional irrigation and fertilization practices that are largely manual and inefficient. Farmers often depend on fixed schedules or visual observation to determine water and nutrient requirements, which does not accurately reflect real-time field conditions. This leads to over-irrigation or under-irrigation, resulting in water wastage, reduced crop yield, and poor soil health. Similarly, improper application of fertilizers causes nutrient imbalance, soil degradation, and environmental pollution due to runoff and leaching. In addition, the absence of real-time monitoring and control systems makes it difficult to maintain optimal growing conditions. These challenges increase labor dependency, operational costs, and the risk of human error. Existing solutions are often expensive, complex, or not easily adaptable for small and medium-scale farmers**[4]**. Therefore, there is a need for a cost-effective, automated, and intelligent system that can precisely control irrigation and nutrient dosing based on real-time data. Such a system should ensure efficient resource utilization, reduce manual effort, and improve overall agricultural productivity while promoting sustainable farming practices**[9]**.

Chapter 2

LITERATURE REVIEW

The study “The Smart Irrigation System Using IoT” (ResearchGate) proposes an automated irrigation framework that uses soil moisture, temperature, and humidity sensors integrated with microcontrollers to optimize water usage . The system enables real-time monitoring and automatic irrigation control, which helps reduce water wastage and improve crop productivity. However, the study is mainly limited to water management and does not include nutrient monitoring or fertigation systems**[1]**.

The paper “IoT-Based Smart Irrigation System” (ResearchGate) presents an IoT-enabled model that allows remote monitoring and control through cloud platforms. It efficiently utilizes sensor data to automate irrigation processes and improve precision agriculture practices. Despite its effectiveness, the system does not incorporate nutrient management techniques, particularly micro- and macro-nutrient delivery mechanisms. **[2]**

In “Smart Irrigation Systems: Overview” (ResearchGate), the authors provide a detailed survey of modern irrigation technologies, focusing on sensor-based and weather-based irrigation systems. The study highlights the importance of au- tomation and IoT in enhancing agricultural efficiency. However, it is limited to irrigation strategies and does not address nutrient-level monitoring or optimization techniques.**[3]**

The review “Smart Drip Irrigation Systems Using IoT: A Review of Architectures, Machine Learning Models and Emerging Trends” discusses advanced irrigation systems that integrate machine learning for predictive decision-making . It emphasizes the role of artificial intelligence in optimizing irrigation schedules and improving crop yield. However, the study does not deeply explore nutrient-specific monitoring or real-time fertigation systems.**[4]**

The research “IoT and Cloud-Based Smart Irrigation System” introduces a cloud-integrated irrigation model designed for large-scale monitoring and data analytics .

The system improves scalability and enables remote accessibility, making it suitable for modern agriculture. However, it focuses mainly on irrigation control and lacks features for nutrient tracking and delivery .**[5]**

The paper "Optimization of Irrigation and Fertigation" (ScienceDirect) addresses both water and nutrient management through optimized fertigation techniques. It demonstrates improved nutrient use efficiency and reduced fertilizer wastage using controlled scheduling. Although it significantly contributes to nutrient management, it does not integrate real-time IoT-based sensing systems.**[6]**

The study "IoT-Enabled Smart Drip Irrigation Using ESP32" (MDPI) presents a cost-effective IoT irrigation system using ESP32. It enables automation and real-time monitoring of environmental conditions. However, the system is limited to irrigation control and does not include advanced nutrient sensing capabilities such as NPK or micronutrient detection.**[7]**

The paper "IoT-Based Irrigation and Monitoring System" (MDPI) introduces a multi-parameter environmental monitoring system that supports automation and better decision-making in agriculture . While it enhances monitoring capabilities, it does not include dedicated nutrient detection modules for soil fertility analysis.**[8]**

The study "Evaluation of IoT-Based Smart Drip Irrigation" (ScienceDirect) evaluates smart irrigation systems based on water efficiency and crop yield improvement . The results indicate significant enhancements in agricultural productivity. However, nutrient optimization is not considered as part of the system design .**[9]**

The paper "IoT-Based Fertigation System for Agriculture" (ResearchGate, 2024) proposes an automated fertigation system using IoT and ESP32 microcontroller . It maintains proper nutrient concentration by controlling electrical conductivity (EC) levels and enables real-time monitoring through an IoT dashboard. The system provides alerts when parameters exceed threshold values and improves efficiency in fertilizer usage and crop growth. However, it primarily focuses on EC control rather than direct measurement of individual nutrients such as nitrogen, phosphorus, and potassium .**[10]**

2.1 Existing System

The existing systems in smart agriculture predominantly focus on automated irrigation using IoT technologies(Figure 2.1) .**[2]** These systems utilize soil moisture sensors, environmental sensors, and cloud-based monitoring to control water supply efficiently. Some advanced systems incorporate machine learning and cloud analytics to improve irrigation scheduling and decision-making.**[7]**
However, most existing systems:

- Focus primarily on water management.
- Lack integration of nutrient monitoring (NPK, micronutrients).
- Do not support automated fertigation systems.
- Provide limited real-time feedback on soil nutrient conditions.

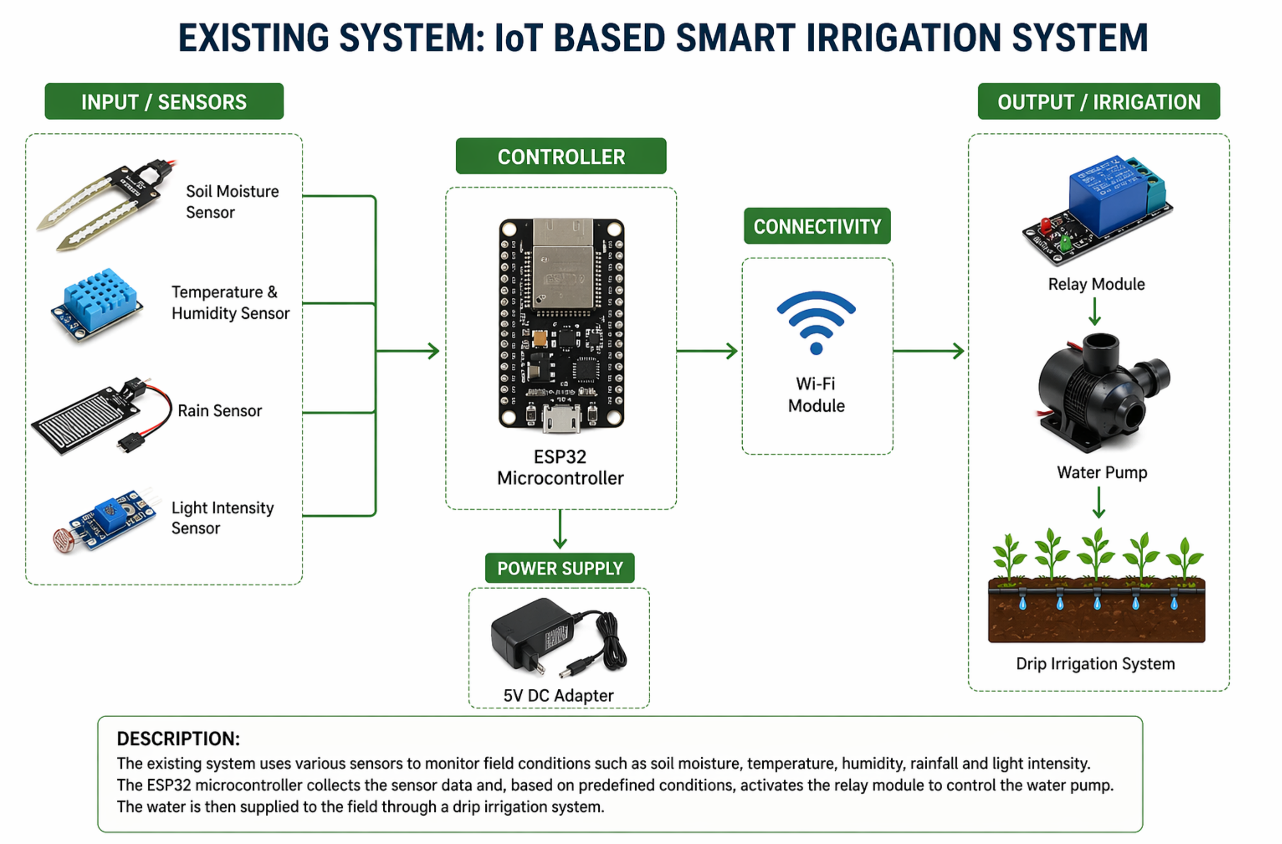

Figure 2.1: **Existing System**

2.2 Related Work

Recent research extends irrigation systems to fertigation by:

- Integrating ESP32-based controllers for automation.
- Using EC sensors for nutrient concentration monitoring.
- Implementing IoT dashboards for real-time control.

These systems improve:

- Fertilizer efficiency
- Crop yield
- Automation level

But still lack:

- Direct NPK sensing
- Micro-nutrient monitoring
- Intelligent nutrient recommendation systems

2.3 Research Gap

- **Indirect Nutrient Measurement:** Most systems rely on EC instead of direct NPK sensing.
- **Lack of Micro-Nutrient Monitoring:** Trace elements (Zn, Fe, Mg) are not considered.
- **No Intelligent Nutrient Decision System:** Systems do not recommend optimal nutrient dosage dynamically.
- **Limited Closed-Loop Fertigation:** Feedback-based nutrient correction is still underdeveloped.

Chapter 3

PROJECT DESCRIPTION

3.1 Existing System

The existing systems in smart agriculture mainly focus on IoT-based automated irrigation techniques that aim to optimize water usage. These systems typically use soil moisture sensors, temperature sensors, and humidity sensors to monitor environmental conditions and control irrigation accordingly.**[1]** The collected data is processed using microcontrollers such as Arduino or ESP32, and in some cases, transmitted to cloud platforms for monitoring and analysis. Advanced systems also incorporate machine learning algorithms to improve irrigation scheduling based on historical and real-time data. These technologies have significantly reduced water wastage and minimized manual effort in agricultural practices, making farming more efficient and sustainable.**[3]**

However, these existing systems have several limitations that affect their overall performance in precision agriculture. One major disadvantage is that they primarily focus only on water management and do not consider nutrient requirements of the plants.**[5]** There is a lack of integration of nutrient monitoring systems, particularly for macro nutrients such as nitrogen, phosphorus, and potassium (NPK), and micro nutrients like zinc, iron, and magnesium.**[7]** Additionally, most systems do not support automated fertigation, which is essential for balanced crop growth. They also rely on indirect measurements such as electrical conductivity (EC) instead of directly measuring nutrient levels. Furthermore, these systems lack real-time feedback mechanisms and intelligent decision-making capabilities, which limits their effectiveness in maintaining soil health and maximizing crop productivity.**[9]**

3.2 Proposed System

The proposed system introduces an advanced IoT-based smart agriculture solution that integrates both irrigation and nutrient management. This system utilizes multiple sensors to monitor soil moisture, temperature, humidity, pH, and electrical conductivity, along with additional sensors for detecting macro and micro nutrient levels.**[1]** The data collected from these sensors is processed using a microcontroller such as ESP32, which enables real-time monitoring and automated control of irrigation and fertigation processes. The system operates in a closed-loop manner, where continuous feedback from sensors is used to adjust water and nutrient supply dynamically based on soil conditions.**[10]**

The proposed system offers several advantages over existing systems. It ensures precise delivery of both water and nutrients, thereby improving nutrient use efficiency and promoting healthy plant growth (Figure 3.1) . The integration of real-time monitoring and automation reduces manual intervention and enhances decision-making accuracy. The system also supports intelligent nutrient recommendations based on soil data, which helps in maintaining optimal soil conditions. Additionally, the use of IoT technology enables scalability and remote monitoring, making the system suitable for different types of agricultural environments. Overall, the proposed system improves crop yield, reduces resource wastage, and provides a comprehensive solution for precision agriculture.**[5]**

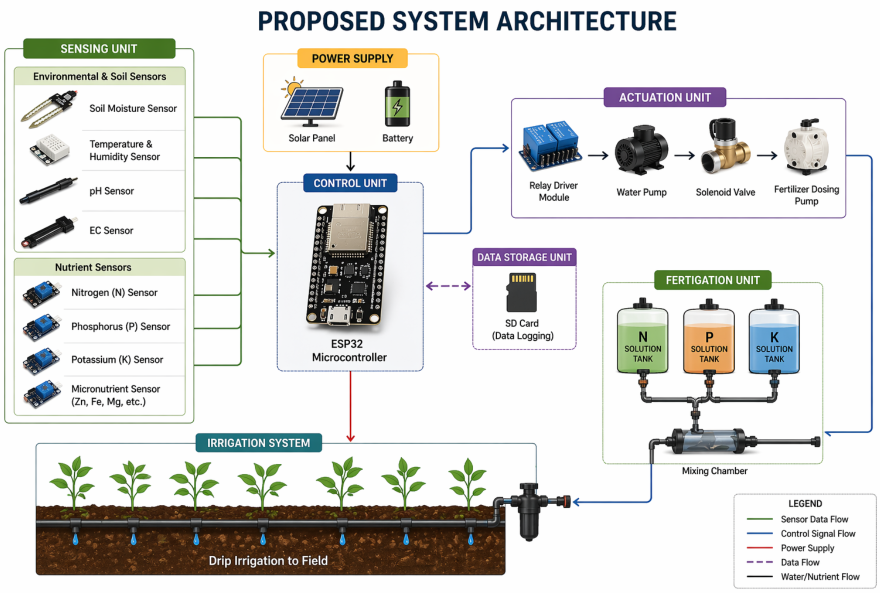

Figure 3.1: **Proposed System**

3.3 Feasibility Study

The feasibility of the proposed smart irrigation and nutrient management system is evaluated based on its practicality, cost-effectiveness, technical requirements, and social impact. The system is designed using readily available hardware components such as sensors, microcontrollers, and communication modules, which makes it practical for real-world agricultural implementation. It can be easily adapted to different soil types and crop requirements, ensuring flexibility and scalability. The integration of IoT technology allows efficient monitoring and control, while automation reduces human effort. The system is capable of improving resource utilization and crop productivity, making it a viable solution for modern agriculture. Overall, the feasibility study indicates that the proposed system is suitable for deployment and offers significant benefits to farmers.

3.3.1 Economic Feasibility

The proposed system is economically feasible as it utilizes low-cost and easily available components such as ESP32 microcontrollers, sensors, and basic communication modules. Although the initial investment may be slightly higher compared to traditional irrigation systems, the long-term benefits justify the cost. The system reduces water wastage and optimizes fertilizer usage, which leads to significant savings in operational costs. Improved crop yield and quality further increase the profitability for farmers. Maintenance costs are minimal, as the components are durable and require less frequent replacement. Additionally, the scalability of the system allows it to be implemented in both small-scale and large-scale farms, making it a cost-effective solution for sustainable agriculture.

3.3.2 Technical Feasibility

The technical feasibility of the proposed system is supported by the availability of modern IoT technologies, sensors, and communication protocols. The ESP32 microcontroller provides sufficient processing capability and built-in Wi-Fi connectivity for real-time data transmission and control. The integration of sensors for soil moisture, pH, electrical conductivity, and nutrients ensures accurate monitoring of environmental and soil conditions. The system can be easily programmed and deployed using existing software tools and platforms. Cloud-based data storage and analytics further enhance the system's functionality by enabling efficient data processing and visualization. Therefore, the proposed system is technically feasible and can be successfully implemented in agricultural applications.

3.3.3 Social Feasibility

The proposed system is socially feasible as it promotes sustainable agricultural practices and improves the overall quality of farming. By automating irrigation and nutrient management, the system reduces the physical workload on farmers and saves time. It also helps in conserving natural resources such as water and soil nutrients, contributing to environmental sustainability. The system can be easily adopted by farmers with basic technical knowledge, as it provides simple and user-friendly operation. Additionally, increased crop yield and reduced costs improve the economic condition of farmers, leading to better livelihoods. The adoption of such smart technologies also encourages modernization in agriculture, making it more efficient and productive.

3.4 System Specification

- Microcontroller: ESP32 (Dual-core processor, Wi-Fi enabled, low power consumption)
- Programming Language: Embedded C
- Development Environment: Arduino IDE (latest version)
- Sensors Used:
  - Soil Moisture Sensor (capacitive type)
  - Temperature and Humidity Sensor (DHT11/DHT22)
  - pH Sensor (for soil acidity/alkalinity)
  - Electrical Conductivity (EC) Sensor (for nutrient estimation)
  - NPK Sensor (for macro nutrient detection)
- Actuators:

  - Relay Module
  - Water Pump
  - Solenoid Valve
  - Fertilizer Dosing Pump

- Power Supply: 5V DC Adapter / Battery / Solar Panel (optional)
- Communication: Wi-Fi module (built-in ESP32)
- Storage: Optional SD Card module for data logging
- System Type: IoT-based Smart Irrigation and Fertigation System

3.4.1 Tools and Technologies Used

Arduino IDE

Arduino IDE is an open-source development environment used for writing, compiling, and uploading code to microcontrollers such as ESP32. It supports programming in Embedded C and provides a simple interface for integrating sensors and actuators. The platform includes various libraries that simplify hardware interfacing and enable rapid prototyping of IoT-based applications. Arduino IDE is compatible with Windows, Linux, and macOS, making it widely accessible for developers and researchers.

Standard Used: ISO/IEC 27001

3.4.2 Standards and Policies

The proposed system follows internationally recognized standards to ensure data security, system reliability, and efficient operation. The primary standard considered is ISO/IEC 27001, which focuses on information security management. This standard ensures that data collected from sensors and transmitted through IoT networks is protected against unauthorized access and breaches. Additionally, the system design adheres to best practices in embedded system development and IoT communication protocols to ensure safe and efficient operation. Proper handling of electrical components and adherence to safety guidelines are also maintained to ensure system reliability and user safety.

Chapter 4

SYSTEM DESIGN AND METHODOLOGY

4.1 System Architecture

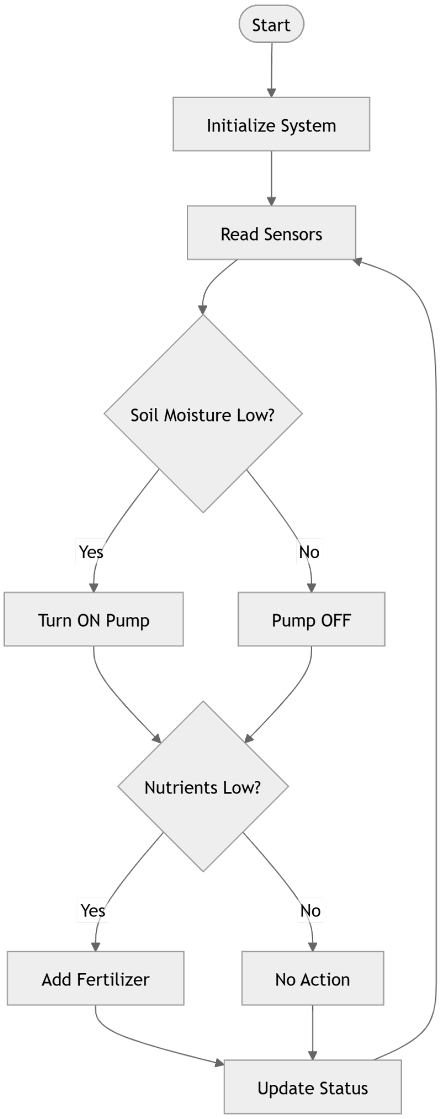

Figure 4.1: **System Architecture**

Figure 4.1 represents the working of the proposed IoT-based smart irrigation and nutrient management system. The process begins with system initialization, where the microcontroller is powered on and all sensors are activated. The system continuously reads data from sensors such as soil moisture and nutrient sensors to monitor field conditions.

Based on the sensor data, the system checks whether the soil moisture level is below the required threshold. If the soil is dry, the water pump is automatically turned ON to supply water; otherwise, the pump remains OFF. After irrigation control, the system evaluates the nutrient levels in the soil. If the nutrient levels are found to be low, the fertilizer pump is activated to provide the required nutrients; otherwise, no action is taken.

Finally, the system updates its status and repeats the process in a continuous loop. This ensures real-time monitoring and automatic control of both irrigation and fertigation, improving efficiency, reducing resource wastage, and supporting healthy plant growth.

4.2 Design Phase

4.2.1 Data Flow Diagram

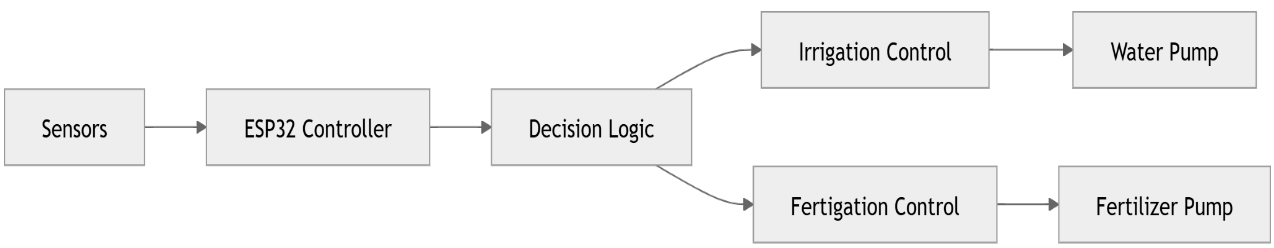

Figure 4.2: **Data Flowchart**

Figure 4.2 illustrates how data moves through the proposed smart irrigation and nutrient management system. Initially, sensors collect real-time data such as soil moisture and nutrient levels from the field. This data is then sent to the ESP32 controller, where it is processed and analyzed. Based on predefined conditions, the controller makes decisions to activate either the irrigation system or the fertigation system. The water pump is turned ON when soil moisture is low, while the fertilizer pump is activated when nutrient levels are insufficient. This continuous flow of data ensures efficient monitoring and automatic control of water and nutrient supply, improving crop growth and reducing resource wastage.

4.2.2 Use Case Diagram

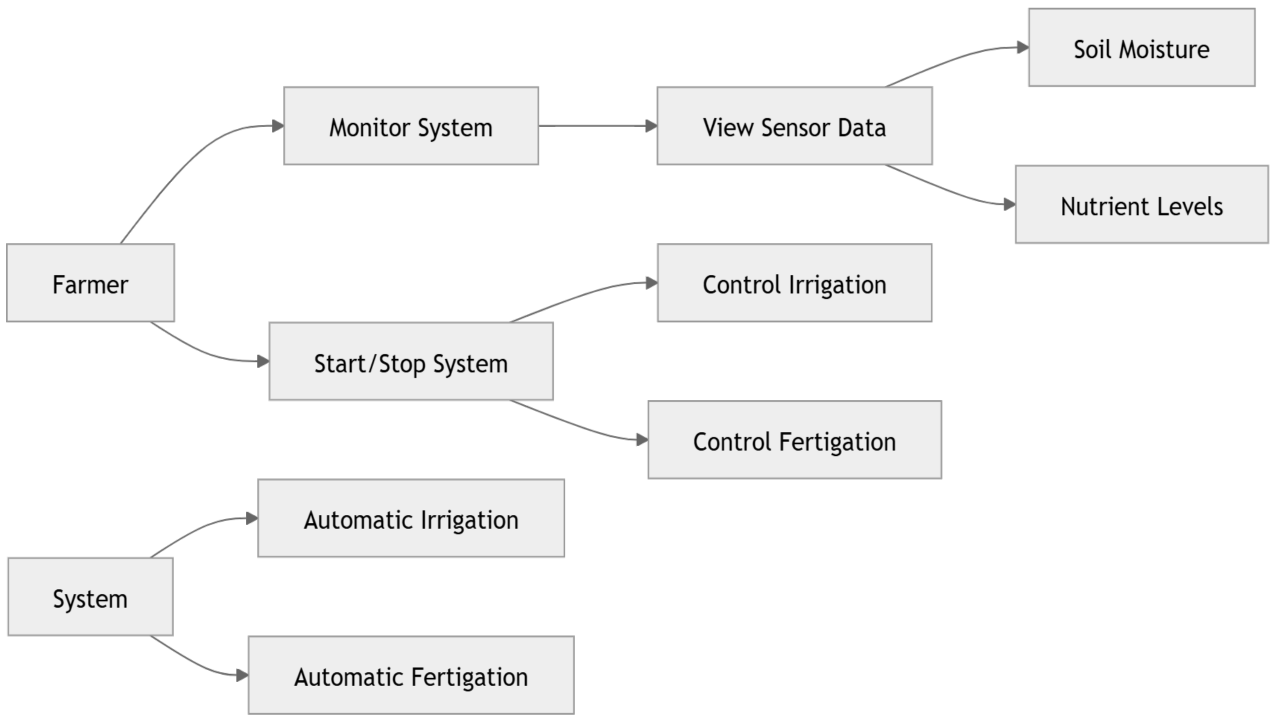

Figure 4.3: **Use Case**

Figure 4.3 represents the interaction between the user (farmer) and the proposed smart irrigation and nutrient management system. The farmer acts as the primary actor who interacts with the system to monitor and control agricultural operations. The system allows the farmer to view real-time sensor data such as soil moisture levels and nutrient status, which helps in understanding the current condition of the field.

In addition to monitoring, the farmer can manually control the irrigation and fertigation processes when required. The system also includes automated functionalities where it independently analyzes sensor data and performs actions such as turning on the water pump when the soil is dry and activating the fertilizer pump when nutrient levels are low. These use cases ensure efficient resource utilization and reduce manual effort. Overall, the diagram highlights both manual and automatic interactions, demonstrating how the system supports smart and precise agricultural management.

4.2.3 Class Diagram

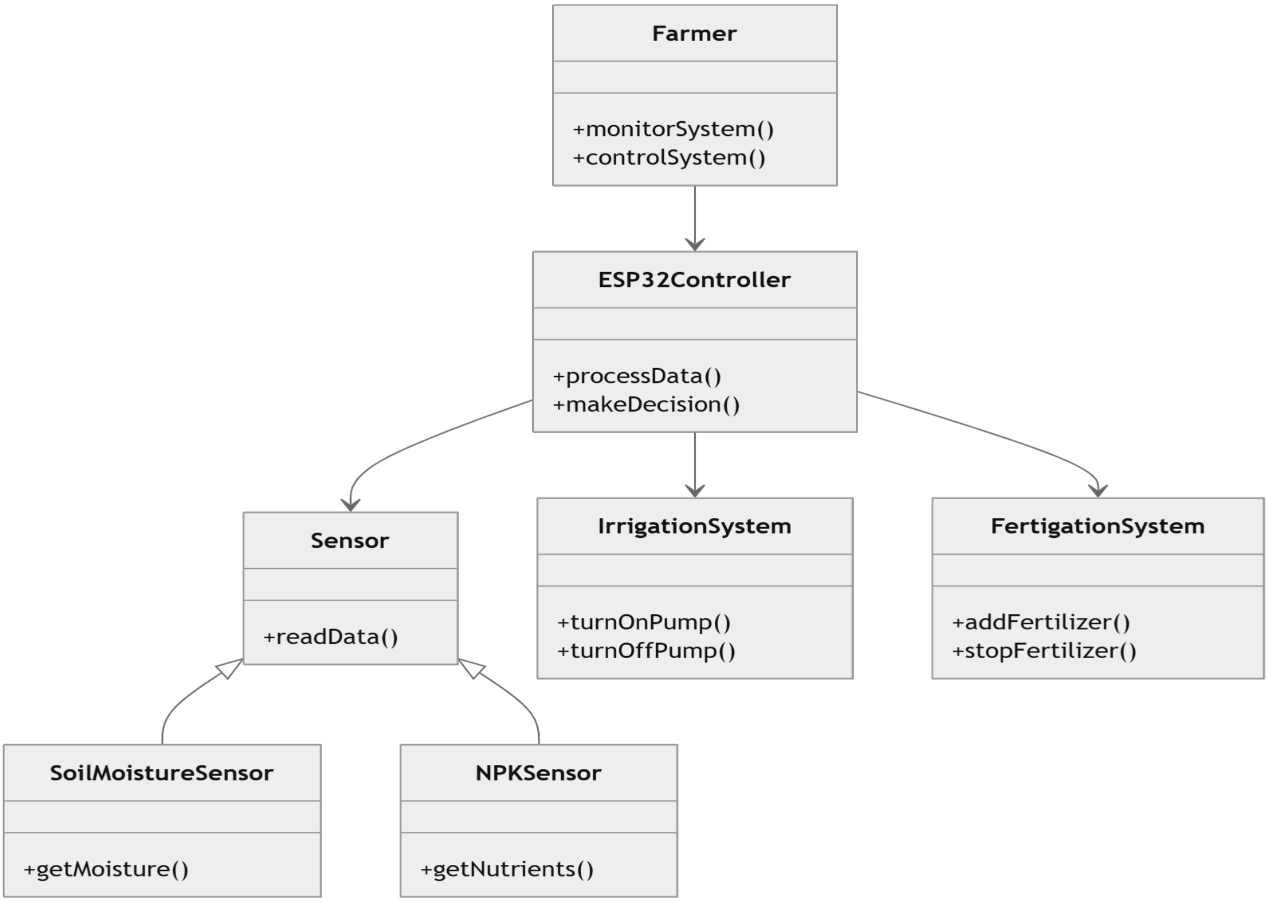

Figure 4.4: **Class Diagram**

Figure 4.4 represents the structure of the proposed smart irrigation and nutrient management system. The Sensor class acts as a base class, with specialized sensors such as SoilMoistureSensor and NPKSensor inheriting its properties and functions. These sensors are responsible for collecting environmental and soil data.

The ESP32Controller class plays a central role by processing the data received from sensors and making decisions based on predefined conditions. Depending on the analysis, it interacts with the IrrigationSystem to control the water pump and the FertigationSystem to manage fertilizer supply. The Farmer class represents the user, who can monitor and control the system when needed. This diagram shows how different components interact to achieve automated and efficient agricultural management.

4.2.4 Sequence Diagram

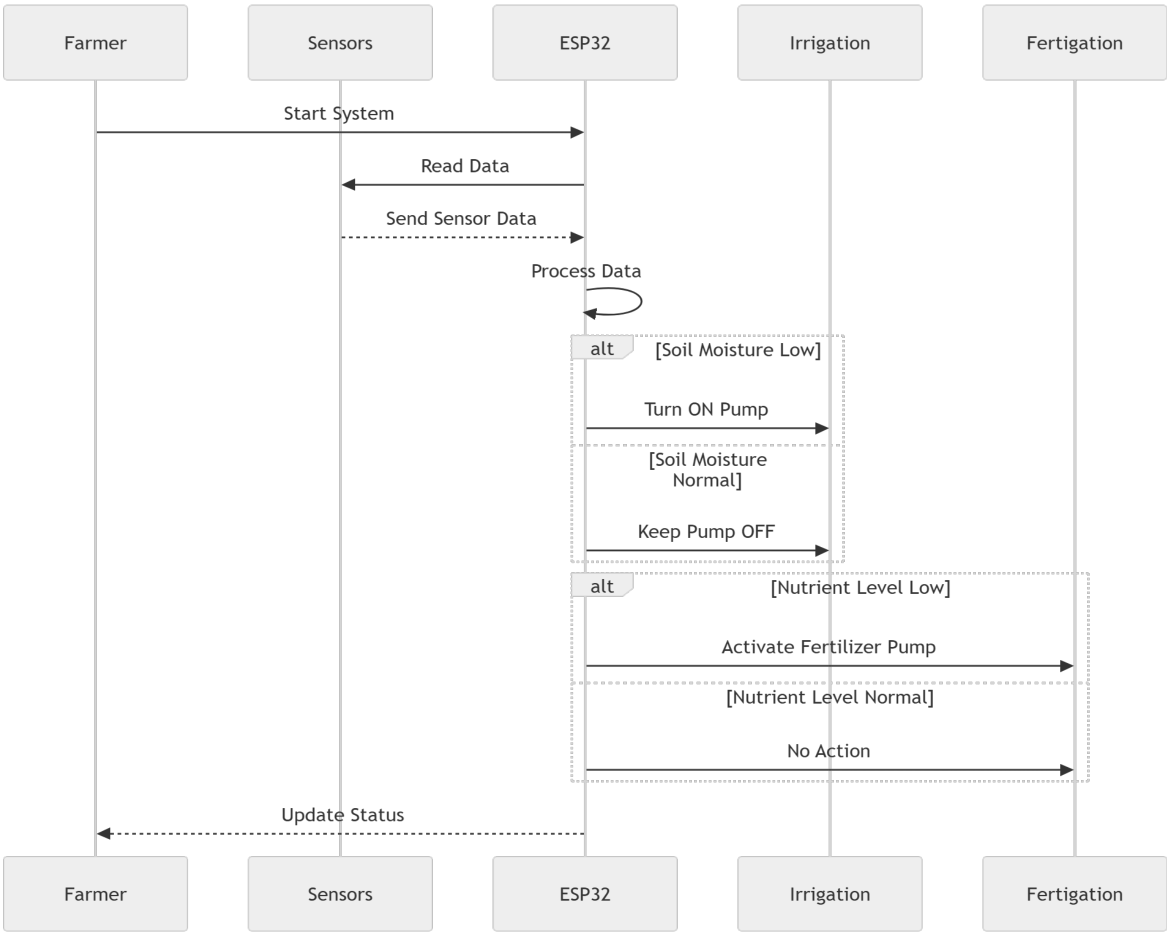

Figure 4.5: **Sequence Diagram**

Figure 4.5 shows the step-by-step interaction between the farmer, sensors, ESP32 controller, and the irrigation and fertigation systems. Initially, the farmer starts the system, after which the ESP32 controller collects data from the sensors. The sensor data is sent back to the controller for processing.

Based on the analysis, the system checks soil moisture levels and decides whether to turn the water pump ON or keep it OFF. Similarly, it evaluates nutrient levels and activates the fertilizer pump if required. Finally, the system updates the status to the farmer. This sequence ensures continuous monitoring and automated decision-making for efficient irrigation and nutrient management.

4.2.5 Collaboration diagram

Figure 4.6: **Collaboration Diagram**

Figure 4.6 represents how different components of the smart irrigation and nutrient management system interact with each other through message exchange. The farmer initiates the system, after which the ESP32 controller communicates with sensors to collect real-time data such as soil moisture and nutrient levels. The sensors send this data back to the controller for processing.

Based on the analysis, the controller sends control signals to the irrigation system and fertigation system. The irrigation system supplies water to the field when required, while the fertigation system provides nutrients. Finally, the controller updates the system status back to the farmer. This diagram emphasizes the communication flow and coordination among system components to ensure efficient operation.

4.2.6 Activity Diagram

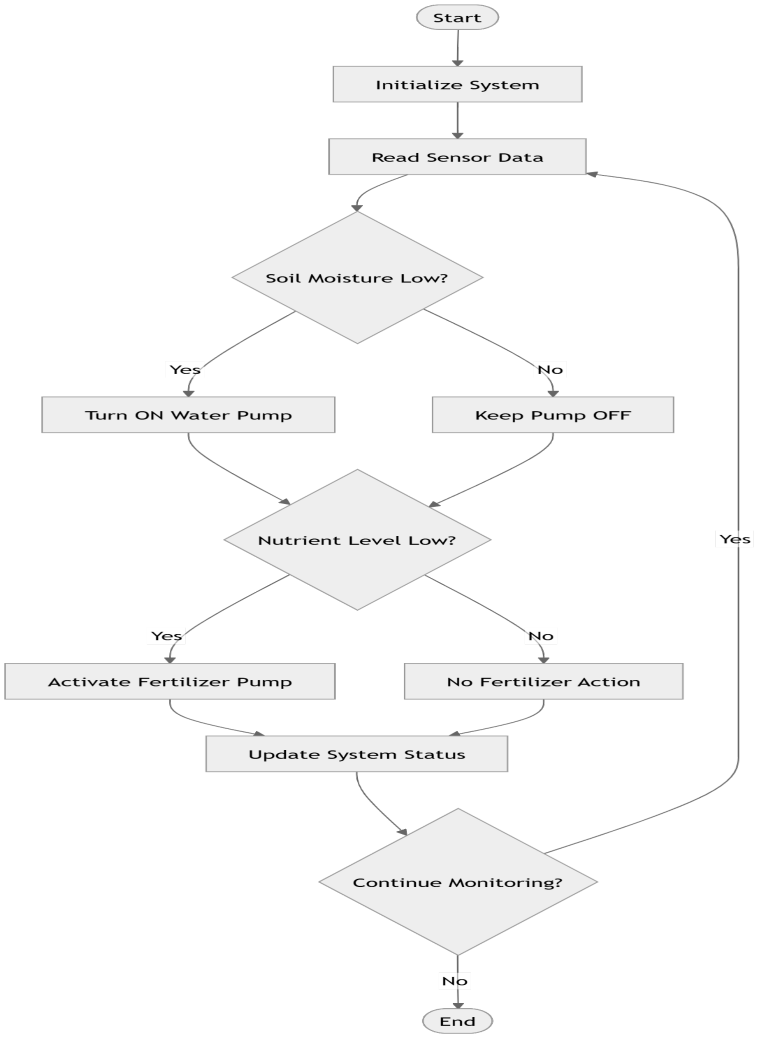

Figure 4.7: **Activity Diagram**

Figure 4.7 illustrates the workflow of the smart irrigation and nutrient management system. The process begins with system initialization, followed by continuous reading of sensor data. The system first checks the soil moisture level and decides whether to turn the water pump ON or keep it OFF. After handling irrigation, it evaluates the nutrient levels in the soil. If nutrients are low, the fertilizer pump is activated; otherwise, no action is taken.

Once both irrigation and fertigation decisions are completed, the system updates its status. The process then continues in a loop to ensure continuous monitoring of environmental conditions. This activity flow ensures efficient use of water and nutrients while maintaining optimal soil conditions for plant growth.

4.3 Algorithm & Pseudo Code

4.3.1 Algorithm

The algorithm for the proposed smart irrigation and nutrient management system is designed to automate the process of monitoring soil conditions and controlling irrigation and fertigation.

1. Start the system.
2. Initialize ESP32 microcontroller and sensors.
3. Read sensor data (soil moisture, temperature, humidity, pH, EC, NPK).
4. Check soil moisture level.
5. If soil moisture is below threshold, turn ON water pump.
6. Else, keep water pump OFF.
7. Check nutrient levels.
8. If nutrient level is low, activate fertilizer pump.
9. Else, no fertilizer action is taken.
10. Update system status.
11. Repeat the process continuously.
12. Stop the system.

4.3.2 Pseudo Code

The pseudo code represents the logical implementation of the system.

```
START
Initialize system
WHILE system is ON DO
    Read soil moisture
    Read nutrient levels
```

```
    IF soil moisture < threshold THEN
        Turn ON water pump
    ELSE
        Turn OFF water pump
    ENDIF

    IF nutrient level < threshold THEN
        Activate fertilizer pump
    ELSE
        Do nothing
    ENDIF

    Update system status
END WHILE
STOP
```

4.4 Module Description

4.4.1 Module1: Sensor Data Acquisition Module

This module is responsible for collecting real-time data from various sensors such as soil moisture, temperature, humidity, pH, and nutrient sensors. The sensors continuously monitor environmental and soil conditions and send the data to the controller. Accurate data acquisition is essential for proper decision-making in the system.

4.4.2 Module2: Data Processing and Control Module

This module processes the sensor data using the ESP32 microcontroller. It analyzes the data and makes decisions based on predefined threshold values. Depending on the analysis, it controls the irrigation system (water pump) and fertigation system (fertilizer pump). This module ensures automation and efficient resource utilization.

4.4.3 Module3: Irrigation and Fertigation Module

This module is responsible for executing actions based on the controller's decisions. It includes components such as relays, water pumps, and fertilizer pumps. The irrigation system supplies water when soil moisture is low, while the fertigation system provides nutrients when required (Figure 4.8) . This module directly affects plant growth and productivity.

4.5 Steps to Execute/Run/Implement the Project

4.5.1 Step1: Hardware Setup

This step involves assembling all hardware components required for the system.

- Connect ESP32 microcontroller to sensors.
- Attach soil moisture, temperature, and nutrient sensors.
- Connect relay module to control pumps.
- Connect water pump and fertilizer pump.
- Ensure proper power supply to all components.

4.5.2 Step2: Software Development

This step involves coding and uploading the program to the microcontroller.

- Open Arduino IDE.
- Write Embedded C program for system logic.
- Include required sensor libraries.

- Compile and verify the code.
- Upload code to ESP32.

4.5.3 Step3: Testing and Execution

This step ensures proper functioning of the system.

- Power ON the system.
- Monitor sensor readings.
- Check automatic operation of water pump.
- Verify fertilizer pump activation.
- Observe system performance and make adjustments if needed.

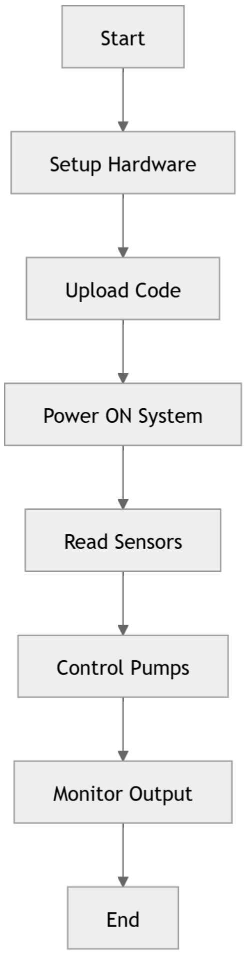

Figure 4.8: **Steps To Execute Project**

4.6 IMPLEMENTATION AND TESTING

4.6.1 Input and Output

The implementation of the proposed smart irrigation and nutrient management system involves handling various inputs from sensors and producing appropriate outputs through actuators. The system continuously collects real-time data from environmental and soil sensors, processes the data using the ESP32 microcontroller, and generates outputs to control irrigation and fertigation systems. This input-output relationship ensures efficient resource utilization and automation in agricultural operations.

Input Design

The input design of the system focuses on collecting accurate and reliable data from multiple sensors. The primary inputs include soil moisture levels, temperature, humidity, pH, electrical conductivity (EC), and nutrient levels (NPK). These inputs are obtained using sensors connected to the ESP32 microcontroller. The sensor data is read at regular intervals and converted into digital signals for processing. Proper calibration of sensors is essential to ensure accuracy and consistency. The input design also considers noise reduction and error handling to maintain data integrity.

Output Design

The output design is responsible for controlling the irrigation and fertigation systems based on the processed input data. The system uses actuators such as relays, water pumps, and fertilizer pumps to deliver water and nutrients to the field. When the soil moisture level falls below a predefined threshold, the system activates the water pump. Similarly, when nutrient levels are low, the fertilizer pump is triggered. The outputs are designed to respond automatically and efficiently, ensuring optimal plant growth while minimizing resource wastage.

4.6.2 Testing

Testing is an essential phase in the implementation of the system to ensure reliability, accuracy, and proper functioning. The system is tested under different environmental conditions to verify its performance. Sensor readings are validated against standard values to ensure correctness. The response of the irrigation and fertigation systems is checked to confirm that they operate according to predefined conditions. Any errors or inconsistencies identified during testing are corrected to improve system performance.

Testing Strategies

Various testing strategies are employed to ensure the effectiveness of the system. Unit testing is performed to verify the functionality of individual components such as sensors and actuators. Integration testing is conducted to ensure proper communication between sensors, controller, and output devices. System testing is carried out to evaluate the overall performance of the system in real-time conditions. Additionally, validation testing is used to confirm that the system meets the project requirements. These testing strategies help in identifying and resolving issues, ensuring a robust and reliable system.

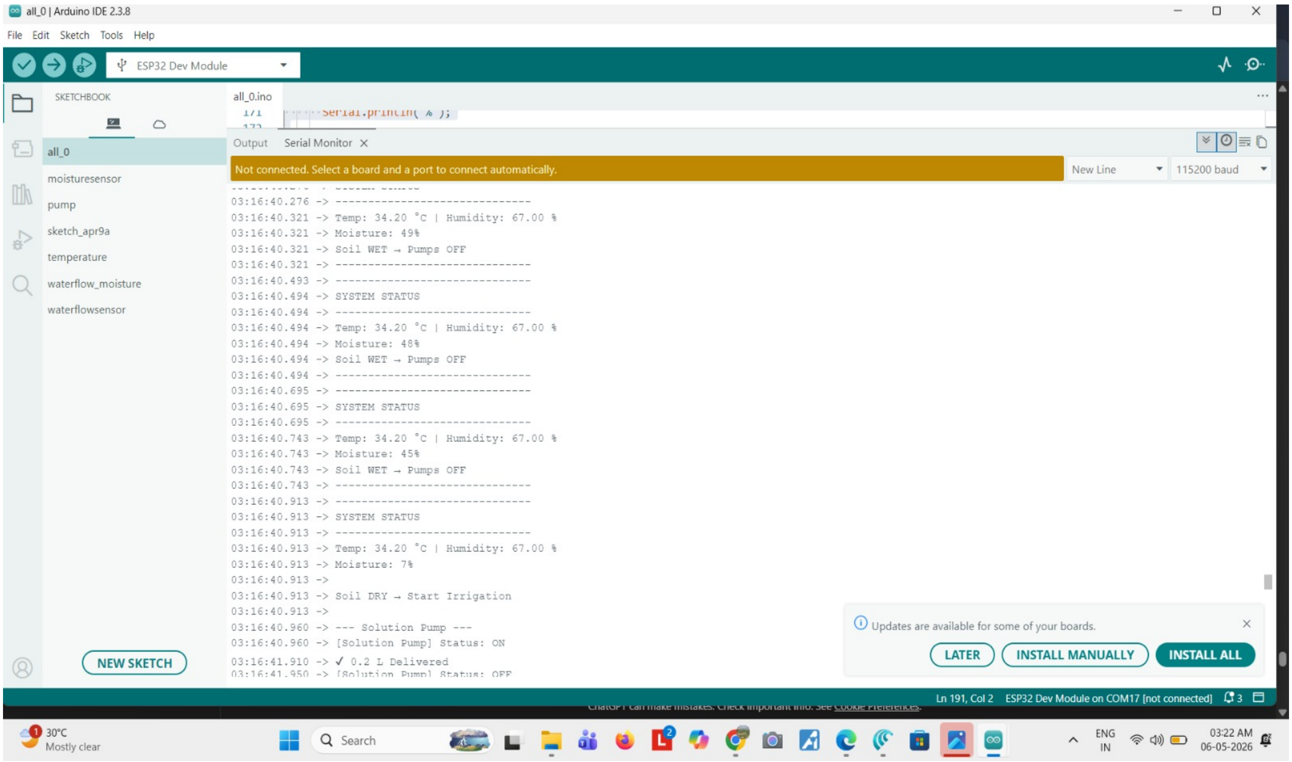

Figure 4.9: **Test Image 1**

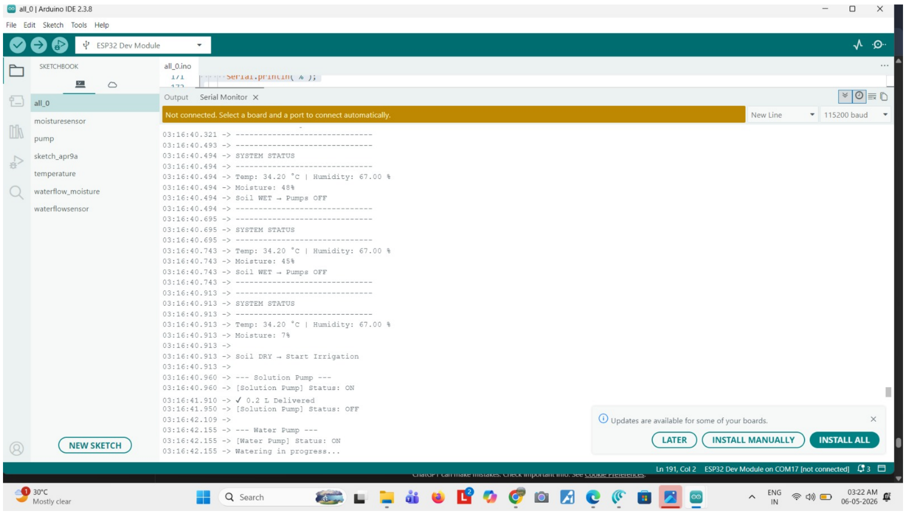

Figure 4.10: **Test Image 2**

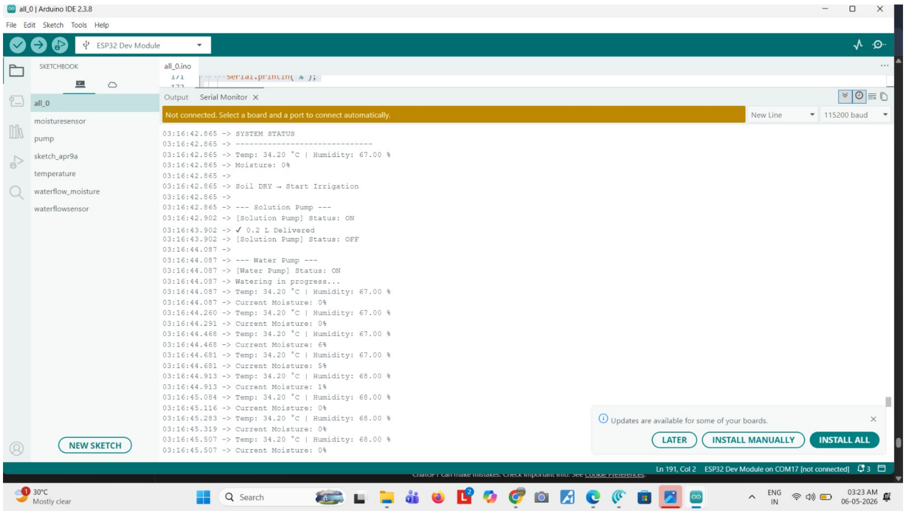

Figure 4.11: **Test Image 3**

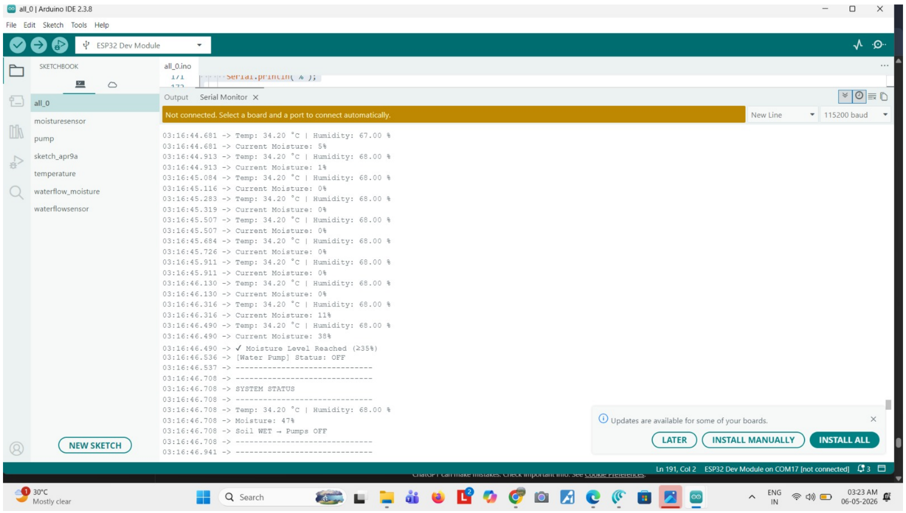

Figure 4.12: **Test Image 4**

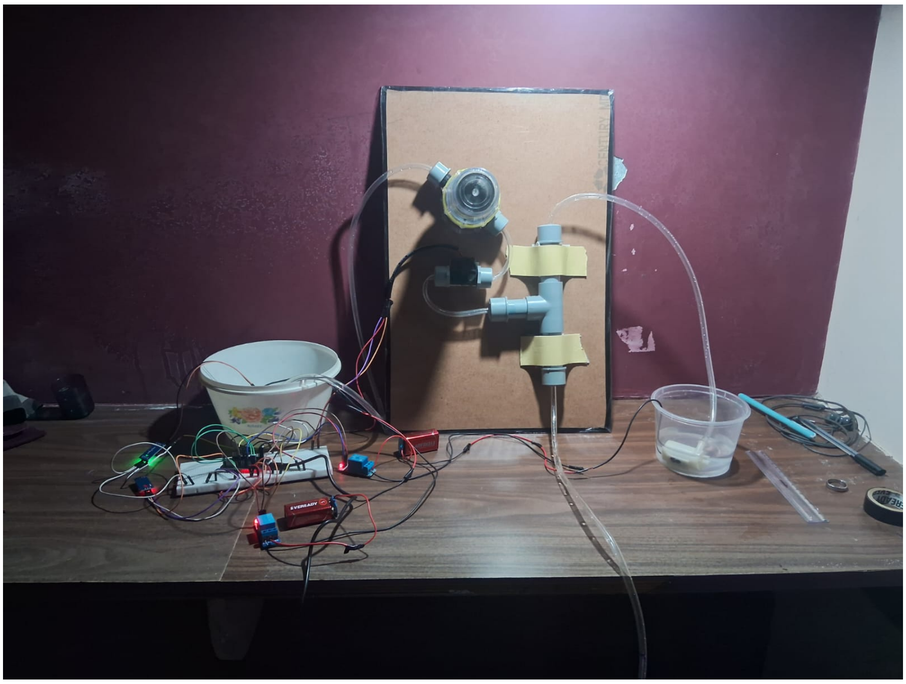

Figure 4.13: **Test Image 5**

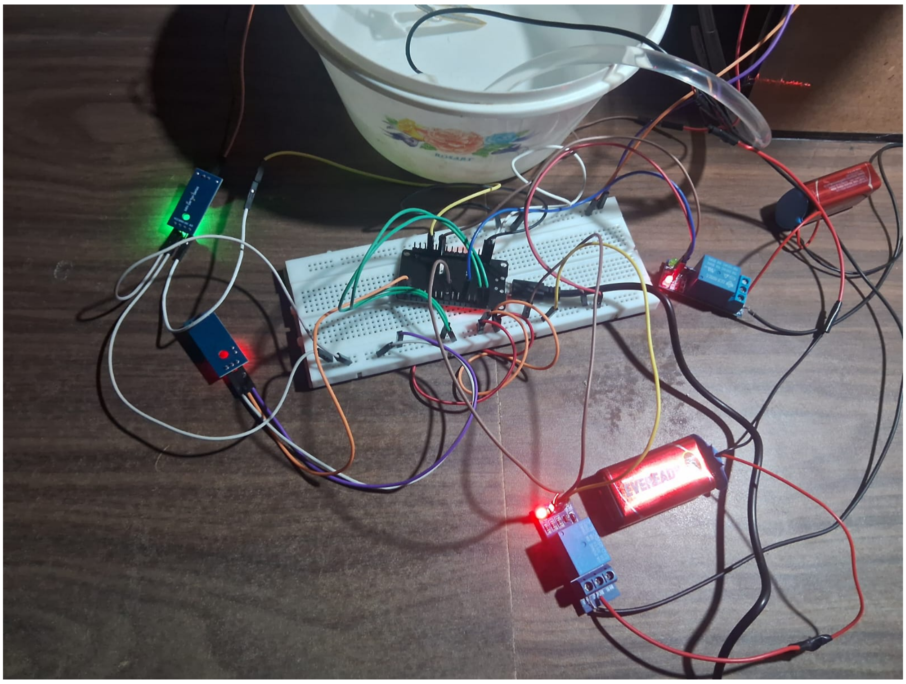

Figure 4.14: **Test Image 6**

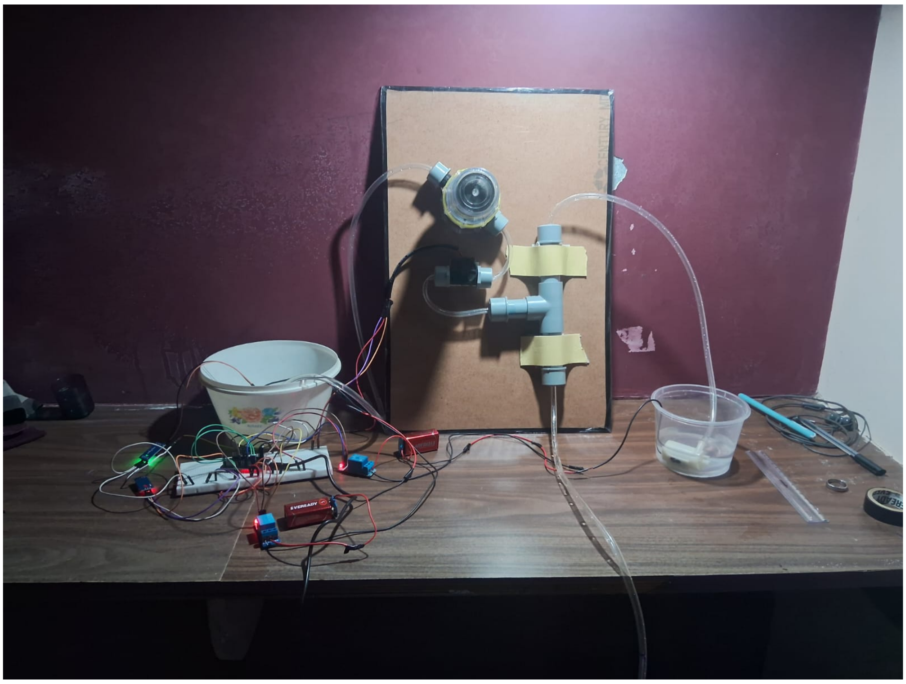

Figure 4.15: **Test Image 7**

Chapter 5

RESULTS AND DISCUSSIONS

5.1 Efficiency of the Proposed System

The efficiency of the proposed smart irrigation and nutrient management system is evaluated based on its ability to optimize water usage and ensure proper nutrient delivery. The system continuously monitors soil conditions using sensors such as soil moisture, pH, and nutrient sensors, and makes real-time decisions to control irrigation and fertigation processes. By automating these operations, the system reduces manual intervention and minimizes human errors. The precise control of water supply prevents over-irrigation and under-irrigation, thereby conserving water resources and improving soil health. Additionally, the integration of nutrient monitoring ensures that plants receive the required macro and micro nutrients at the right time, which enhances overall crop growth and productivity.

The performance of the system also demonstrates improved efficiency in terms of resource utilization and cost-effectiveness. The automated control mechanism reduces unnecessary usage of fertilizers and water, leading to significant savings for farmers. The system operates in a continuous loop, providing real-time feedback and adjustments based on changing environmental conditions. This dynamic response improves the reliability and adaptability of the system in different agricultural scenarios. Furthermore, the use of IoT technology allows seamless monitoring and control, making the system scalable and suitable for both small and large farms. Overall, the proposed system achieves higher efficiency compared to traditional irrigation methods by ensuring optimal resource management and sustainable agricultural practices.

5.2 Comparison of Existing and Proposed System

The comparison between the existing system and the proposed system highlights the improvements achieved through the integration of IoT-based irrigation and nutrient management. The existing systems primarily focus on automated irrigation using soil moisture sensors and basic environmental parameters. However, they lack proper nutrient monitoring and intelligent decision-making capabilities. In contrast, the proposed system integrates both irrigation and fertigation, enabling precise control over water and nutrient supply based on real-time soil conditions.

The proposed system provides enhanced efficiency, automation, and accuracy compared to traditional and existing smart irrigation systems. It utilizes advanced sensors to monitor macro and micro nutrient levels along with environmental parameters. The system operates in a closed-loop manner, ensuring continuous feedback and dynamic adjustment of resources. This results in improved crop yield, reduced resource wastage, and better soil health. The following table summarizes the key differences between the existing and proposed systems (Table 5.1).

Feature	Existing System	Proposed System
Water Management	Basic automated irrigation	Intelligent and optimized irrigation
Nutrient Monitoring	Not available or indirect (EC-based)	Direct monitoring of NPK and micronutrients
Fertigation	Not supported	Fully automated fertigation system
Decision Making	Limited or manual	Automated and real-time decision making
Feedback Mechanism	Open-loop system	Closed-loop system with continuous feedback
Efficiency	Moderate	High efficiency with optimized resource usage
Crop Yield	Limited improvement	Significant improvement in crop productivity
Resource Utilization	Higher wastage	Reduced water and fertilizer wastage

Table 5.1: **Comparison of Existing and Proposed System**

5.3 Comparative Analysis-Table

Feature	Existing System	Proposed System
Water Management	Basic automated irrigation	Intelligent and optimized irrigation
Nutrient Monitoring	Not available or indirect (EC-based)	Direct monitoring of NPK and micronutrients
Fertigation	Not supported	Fully automated fertigation system
Decision Making	Limited or manual	Automated and real-time decision making
Feedback Mechanism	Open-loop system	Closed-loop system with continuous feedback
Efficiency	Moderate	High efficiency with optimized resource usage
Crop Yield	Limited improvement	Significant improvement in crop productivity
Resource Utilization	Higher wastage	Reduced water and fertilizer wastage

Table 5.2: **Comparative Analysis of Existing and Proposed System**

5.4 Comparative Analysis-Graphical Representation and Discussion

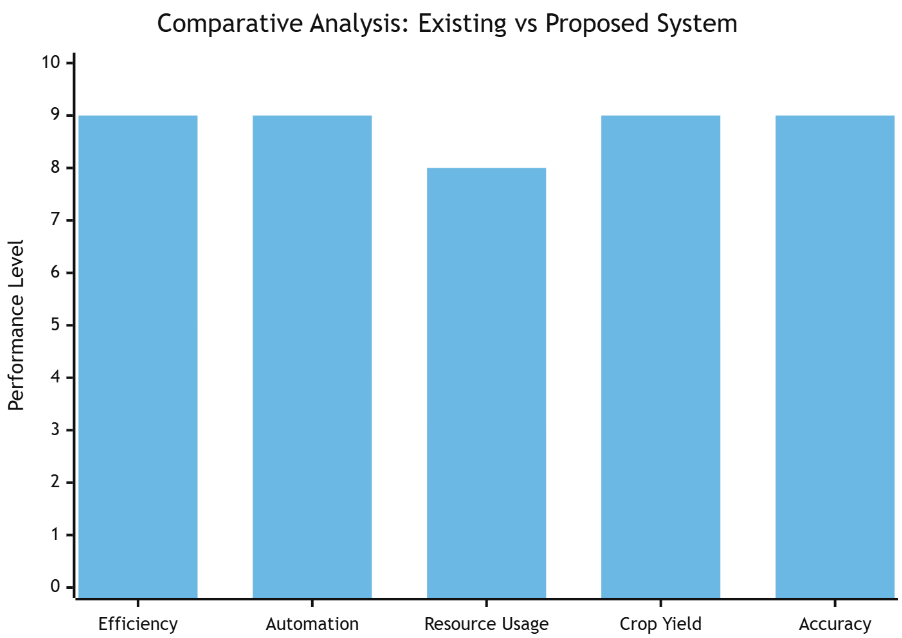

Figure 5.1: **Comparitive Analysis Graph**

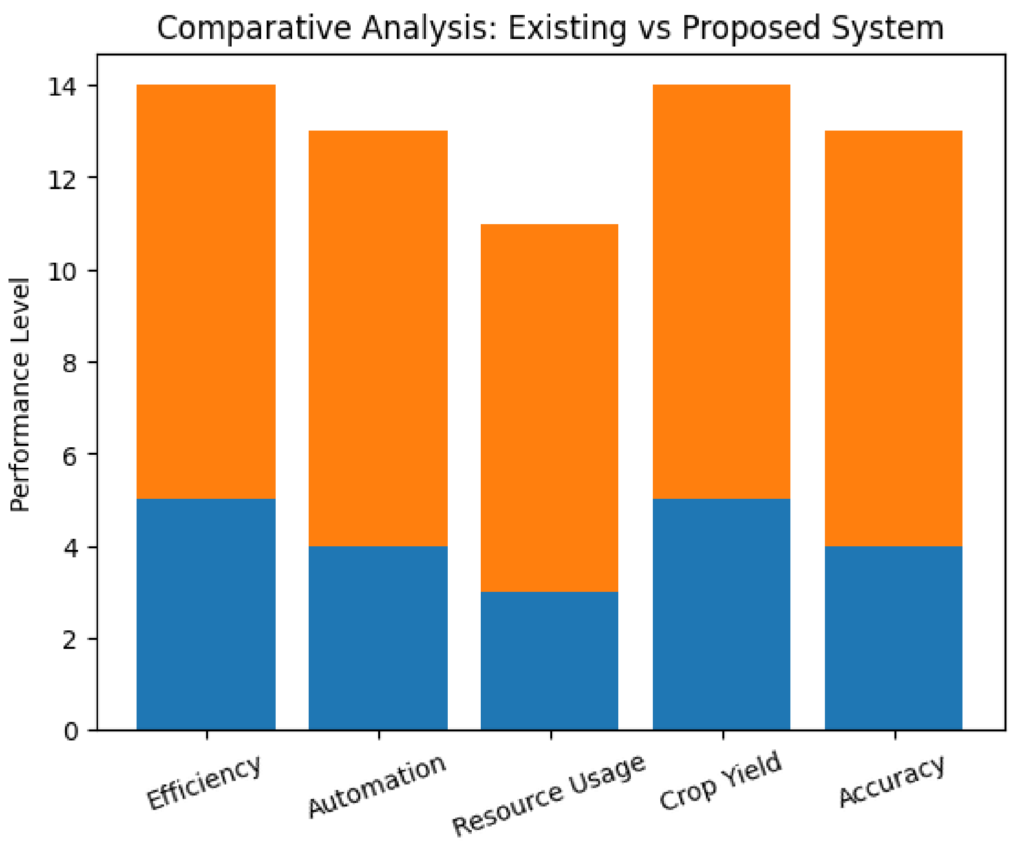

Figure 5.2: **Existing Vs Proposed System**

The graphical representation clearly shows the performance difference between the existing and proposed systems. The existing system demonstrates moderate efficiency and limited automation, primarily focusing on basic irrigation without proper nutrient integration (Figure 5.1). It also shows lower performance in terms of accuracy and resource utilization due to the absence of real-time feedback and intelligent decision-making.

In contrast, the proposed system significantly outperforms the existing system across all parameters. It achieves higher efficiency through precise irrigation and fertigation control, improved automation using IoT-based monitoring, and better resource utilization by minimizing water and fertilizer wastage. The system also enhances crop yield and accuracy by continuously analyzing soil conditions and adjusting operations accordingly. Overall, the graphical analysis highlights the superiority of the proposed system in enabling smart and sustainable agriculture(Figure 5.2).

Chapter 6

CONCLUSION AND FUTURE ENHANCEMENTS

6.1 Summary

This IoT-based smart nutrient monitoring system is designed to enhance efficient resource management in agriculture by continuously monitoring soil and environmental conditions such as soil moisture, temperature, humidity, and nutrient levels. The system uses IoT-enabled sensors connected to a microcontroller (such as Arduino or ESP32) to collect real-time data from the field . This data is transmitted to a cloud platform or mobile application, allowing users to remotely monitor plant health and soil conditions anytime and anywhere.

The proposed system helps automate decision-making by triggering alerts or controlling irrigation and nutrient supply when required thresholds are reached. This reduces manual effort, prevents over-irrigation or under-fertilization, and improves crop productivity. By integrating smart sensing and automation, the project promotes sustainable agriculture practices and efficient utilization of water and nutrients, making the farming process more intelligent, cost-effective, and resource-efficient.

6.2 Limitations

The proposed IoT-based smart nutrient monitoring system has several limitations that may affect its performance and scalability in real-world applications (Figure 6.1) . These limitations are listed below:

- **High Initial Cost:** The system requires multiple sensors, microcontrollers, and cloud services, which increases the initial setup cost.
- **Internet Dependency:** Continuous internet connectivity is required for real-

time data monitoring and cloud communication, which may be unreliable in rural areas.

- **Sensor Accuracy Issues:** Sensor readings may be affected by environmental conditions, calibration errors, or long-term usage degradation.
- **Power Supply Constraints:** The system requires a stable power source, and power failures can interrupt monitoring unless backup systems are used.
- **Maintenance Requirements:** Regular calibration and maintenance of sensors and hardware are necessary for accurate performance.
- **Limited Coverage Area:** A single IoT setup may not be sufficient for large agricultural fields, requiring multiple sensor nodes.
- **Data Security Risks:** Since data is transmitted over the internet or cloud platforms, there is a risk of unauthorized access or data breaches.

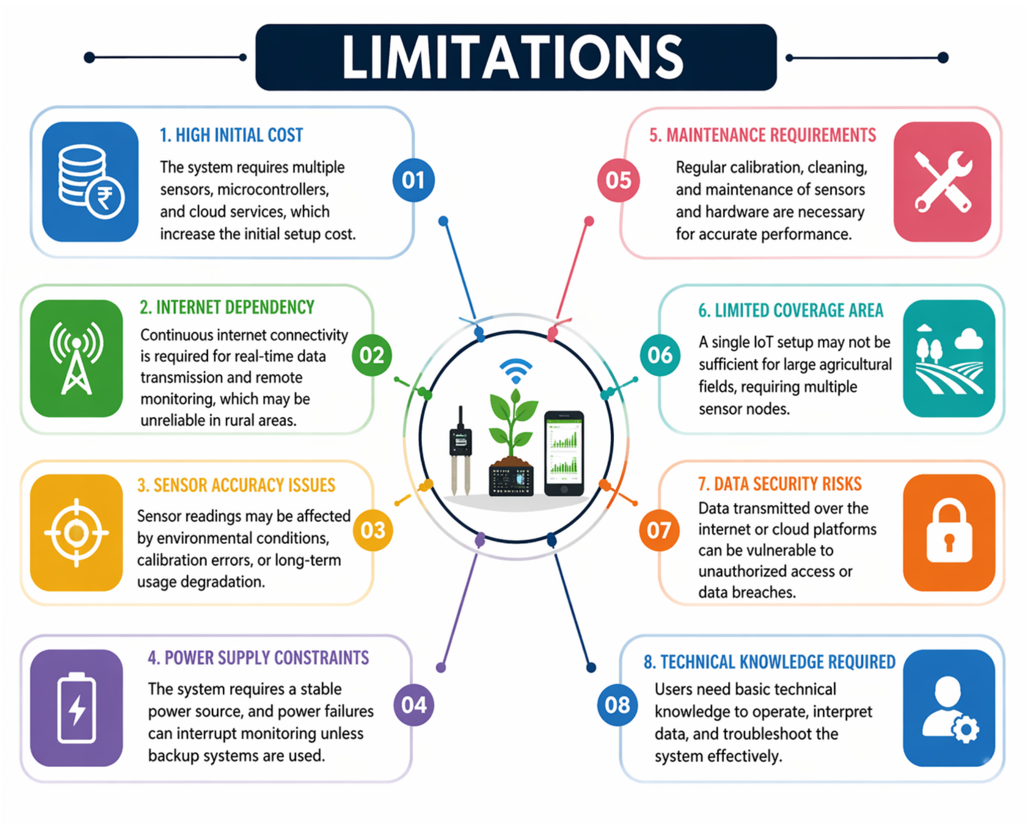

Figure 6.1: **Limitations**

6.3 Future Enhancements

The IoT-based smart nutrient monitoring system can be further improved by incorporating advanced technologies and features to increase its efficiency, scalability, and usability. The following enhancements can be considered for future development:

- **AI-Based Crop Recommendation System:** Integration of machine learning models can help in predicting suitable crops and recommending fertilizers based on soil conditions and historical data.
- **Advanced Sensor Integration:** High-precision sensors such as NPK sensors, soil EC (Electrical Conductivity) sensors, and weather prediction modules can be added for more accurate monitoring.
- **Solar-Powered System:** Using solar energy can make the system more energy-efficient and suitable for remote agricultural areas with limited electricity access.
- **Edge Computing Implementation:** Local data processing using edge devices can reduce dependency on cloud services and improve system response time.
- **Mobile Application Enhancement:** A more advanced mobile application with dashboards, graphical analytics, historical trends, and intelligent alerts can improve user experience.
- **Automated Irrigation and Fertilizer Control:** Integration of pumps and valves can enable full automation of irrigation and nutrient supply based on sensor data.
- **Cloud Data Analytics and Big Data Integration:** Data collected from multiple farms can be analyzed to generate agricultural insights and improve decision-making at a larger scale.
- **IoT Network Expansion (LoRa/5G):** Long-range communication technologies such as LoRa or 5G can improve connectivity in large-scale farming environments.

Chapter 7

ADDRESSING COMPLEX ENGINEERING PROBLEM AND SDG

Level	Attribute	Characteristics	Targeted WKs	Justification
WP1	Depth of Knowledge	Requires in-depth engineering knowledge (K3–K8)	WK3, WK4, WK5, WK6, WK8	The IoT-based smart nutrient monitoring system requires knowledge of embedded systems, sensors, communication networks, and cloud platforms to monitor soil and environmental parameters effectively.
WP2	Range of conflicting requirements	Involves technical and non-technical conflicts	WK4, WK5, WK6, WK8	The system balances cost vs accuracy, power consumption vs performance, and real-time monitoring vs network dependency while ensuring usability for farmers.
WP3	Depth of analysis required	Requires originality and analytical thinking	WK3, WK4, WK5, WK8	The project involves analyzing sensor data, calibrating devices, and designing efficient decision-making systems for irrigation and nutrient control.
WP4	Familiarity of issues	Includes new and unfamiliar challenges	WK4, WK8	Issues such as sensor errors, environmental variations, and rural connectivity problems require innovative handling.
WP5	Extent of applicable codes	Beyond standard engineering practices	WK6, WK8	The system integrates IoT protocols and agricultural practices that require customized solutions beyond standard guidelines.
WP6	Stakeholder involvement	Multiple users involved	WK5, WK6, WK7	Farmers, researchers, and developers are involved, each having different needs such as ease of use, accuracy, and system reliability.
WP7	Interdependence	System-level integration required	WK3, WK5, WK6	The system combines sensors, controllers, cloud platforms, and user interfaces, all working together for efficient operation.

Table 7.1: **Mapping of Project Attributes**

7.1 Mapping of the Project to Complex Engineering Activities (CEA)

EA No.	Attribute	Characteristics	Justification based on the Project
EA1	Range of Resources	Diverse resources and technologies	The project utilizes IoT sensors (soil moisture, temperature, humidity), microcontrollers (Arduino/ESP32), cloud platforms, and wireless communication technologies to monitor and manage agricultural parameters in real time.
EA2	Level of Interactions	Optimization of complex interactions	The system coordinates interactions between sensors, microcontroller, cloud storage, and user interfaces to ensure efficient data collection, processing, and real-time monitoring.
EA3	Innovation	Creative and research-based solutions	The project introduces an IoT-based smart monitoring system that enables automated irrigation and nutrient management using real-time sensor data, improving efficiency and decision-making in agriculture.
EA4	Consequences to Society and Environment	Significant societal impact	The system supports sustainable agriculture by reducing water wastage, optimizing fertilizer usage, improving crop yield, and minimizing environmental impact.
EA5	Familiarity	Extends beyond prior experience	The project involves real-time IoT implementation, sensor calibration, and integration challenges that go beyond traditional theoretical knowledge, requiring practical problem-solving skills.

Table 7.2: **Mapping of Project to Complex Engineering Activities (CEA)**

7.2 SDG Mapping (CEP Section)

SDG No.	SDG Title	Relevance to the Project
SDG 2	Zero Hunger	The project improves agricultural productivity by monitoring soil nutrients and environmental conditions, helping farmers make better decisions to increase crop yield.
SDG 6	Clean Water and Sanitation	Optimizes water usage through smart irrigation, reducing water wastage and promoting efficient water management in agriculture.
SDG 9	Industry, Innovation and Infrastructure	Encourages the adoption of IoT technologies in agriculture, promoting innovation and development of smart farming infrastructure.
SDG 12	Responsible Consumption and Production	Reduces excessive use of fertilizers and resources by providing accurate data for efficient nutrient management.
SDG 13	Climate Action	Supports sustainable farming practices by minimizing environmental impact and promoting efficient resource utilization.

Table 7.3: **SDG Mapping for the Project**

Chapter 8

PLAGIARISM REPORT

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DESIGN AND DEVELOPMENT OF SMART PLANT MICRO-MACRO NUTRIENT DOSER FOR DRIP IRRIGATION

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Your text contains similarities to the following sources:

Smart nutrient dosing in drip irrigation plays a crucial role in precision agriculture. The system integrates sensors, IoT and automated control to deliver exact amounts of macro and micro nutrients. This helps in optimizing plant growth and reducing nutrient wastage. 1

Micro and macro nutrients are essential for plant development. Nitrogen, phosphorus, potassium and secondary nutrients influence yield and quality. Automated dosing systems ensure correct concentration in irrigation water. 2

IoT based monitoring enables real-time measurement of parameters like EC, pH, temperature and flow. Data is transmitted to cloud for analysis and decision making. 3

Researchers have developed various fertigation systems using sensors and controllers. These systems improved nutrient use efficiency and crop productivity under drip irrigation. 4

Automation in fertigation reduces manual effort and improves accuracy. The proposed system uses IoT, sensors and dosing pumps to maintain required nutrient levels. 5

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May 4, 2026

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Figure 8.1: **Plagarism Report**

Chapter 9

SOURCE CODE

9.1 Source Code

```
#include <Adafruit_Sensor.h>
#include <DHT.h>

// ------------------- Pin Definitions -------------------
#define DHTPIN 23
#define DHTTYPE DHT11

#define MOISTURE_ANALOG 34
#define FLOW_SENSOR_PIN 32

#define PUMP1_RELAY 4    // Water Pump
#define PUMP2_RELAY 5    // Solution Pump

// ------------------- Objects -------------------
DHT dht(DHTPIN, DHTTYPE);

// ------------------- Flow Sensor Variables -------------------
volatile int pulseCount = 0;
float flowRate = 0.0;
float totalLitres = 0.0;
unsigned long lastTime = 0;

const float calibrationFactor = 7.5;

// ------------------- Thresholds -------------------
const float SOLUTION_LIMIT = 0.2;
const int MOISTURE_THRESHOLD = 35;

// ------------------- Interrupt -------------------
void IRAM_ATTR countPulse() {
  pulseCount++;
}

// ------------------- Helper Functions -------------------
void printLine() {
  Serial.println("------------------------------");
```

```
}

void printPumpStatus(String pumpName, String status) {
  Serial.print("[");
  Serial.print(pumpName);
  Serial.print("] Status: ");
  Serial.println(status);
}

void printDHT() {
  float humidity = dht.readHumidity();
  float temperature = dht.readTemperature();

  if (isnan(humidity) || isnan(temperature)) {
    Serial.println("DHT Error!");
    return;
  }

  Serial.print("Temp: ");
  Serial.print(temperature);
  Serial.print(" C | Humidity: ");
  Serial.print(humidity);
  Serial.println(" %");
}

// ------------------- Setup -------------------
void setup() {
  Serial.begin(115200);

  dht.begin();
  analogReadResolution(12);

  pinMode(FLOW_SENSOR_PIN, INPUT);

  pinMode(PUMP1_RELAY, OUTPUT);
  pinMode(PUMP2_RELAY, OUTPUT);

  digitalWrite(PUMP1_RELAY, HIGH);
  digitalWrite(PUMP2_RELAY, HIGH);

  attachInterrupt(digitalPinToInterrupt(FLOW_SENSOR_PIN), countPulse, RISING);

  Serial.println("System Ready...");
}

// ------------------- Loop -------------------
void loop() {

```

```
int analogValue = analogRead(MOISTURE_ANALOG);
int moisturePercent = map(analogValue, 4095, 0, 0, 100);
moisturePercent = constrain(moisturePercent, 0, 100);

printLine();
Serial.println("SYSTEM STATUS");
printLine();

printDHT();

Serial.print("Moisture: ");
Serial.print(moisturePercent);
Serial.println("%");

bool isDry = (moisturePercent < MOISTURE_THRESHOLD);

if (isDry) {

  Serial.println("\nSoil DRY      Start Irrigation");

  // ================ SOLUTION PUMP ================
  Serial.println("\n--- Solution Pump ---");
  printPumpStatus("Solution Pump", "ON");

  digitalWrite(PUMP2_RELAY, LOW);
  digitalWrite(PUMP1_RELAY, HIGH);

  totalLitres = 0;
  pulseCount = 0;
  lastTime = millis();

  unsigned long startTime = millis();

  while (totalLitres < SOLUTION_LIMIT) {

    if (millis() - startTime > 15000) {
      Serial.println("Timeout: Flow issue");
      break;
    }

    if (millis() - lastTime >= 1000) {

      noInterrupts();
      int pulses = pulseCount;
      pulseCount = 0;
      interrupts();

      flowRate = pulses / calibrationFactor;
```

```
        float litresPerSecond = flowRate / 60.0;
        totalLitres += litresPerSecond;

        lastTime = millis();
      }
    }

    Serial.println("0.2 L Delivered");

    digitalWrite(PUMP2_RELAY, HIGH);
    printPumpStatus("Solution Pump", "OFF");

    delay(200);

    // ================ WATER PUMP ================
    Serial.println("\n--- Water Pump ---");
    printPumpStatus("Water Pump", "ON");
    Serial.println("Watering in progress...");

    digitalWrite(PUMP1_RELAY, LOW);

    startTime = millis();

    while (true) {

      if (millis() - startTime > 20000) {
        Serial.println("Timeout: Moisture not increasing");
        break;
      }

      printDHT();

      int analogValue = analogRead(MOISTURE_ANALOG);
      int moisturePercent = map(analogValue, 4095, 0, 0, 100);
      moisturePercent = constrain(moisturePercent, 0, 100);

      Serial.print("Current Moisture: ");
      Serial.print(moisturePercent);
      Serial.println("%");

      if (moisturePercent >= MOISTURE_THRESHOLD) {
        Serial.println("Moisture Level Reached (>=35%)");
        digitalWrite(PUMP1_RELAY, HIGH);
        printPumpStatus("Water Pump", "OFF");
        break;
      }

      delay(200);
```

```
    }

  } else {
    digitalWrite(PUMP1_RELAY, HIGH);
    digitalWrite(PUMP2_RELAY, HIGH);
    Serial.println("Soil WET    Pumps OFF");
  }

  printLine();
  delay(200);
}
```

Chapter 10

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